Spectral Lower Bound and an Improved Upper Bound for the Chromatic Number of Unit Sphere Packings

Abstract

A *unit sphere packing* in $\mathbb{R}^n$ consists of unit spheres of radius $\frac{1}{2}$ with pairwise disjoint interiors, equivalently a set of centres with pairwise distance ≥ 1. The *tangency graph* connects pairs of centres at distance exactly 1, and the *chromatic number* of a packing is the chromatic number of its tangency graph. Let $M(n)$ denote the supremum of chromatic numbers of unit sphere packings in $\mathbb{R}^n$.

Lower bound. We give a self-contained spectral proof that

$$\limsup_{n\to\infty} \frac{M(n)-n}{n^{2/3}} \;\geq\; 1,$$

and more precisely $M(d_q) \geq q^3+1 = d_q + (q^2-q+1)$ for every prime power $q \geq 2$, where $d_q = q^3 - q^2 + q$. The proof combines: (i) a *Spectral Realization Theorem* showing that any connected k-regular graph with least adjacency eigenvalue $s < -1$ of multiplicity g embeds as the tangency graph of a unit sphere packing in $\mathbb{R}^{v-g-1}$, and (ii) the classical *Hoffman ratio bound* $\chi(G) \geq 1 - k/s$. The witness is the complement of the collinearity graph of $\mathrm{GQ}(q, q^2)$.

Upper bound. We prove an improved upper bound

$$M(n) \;\leq\; O\!\left(\tau(n)\cdot\sqrt{\frac{\log n}{n}}\right) \;=\; 2^{0.401n(1+o(1))}\cdot O\!\left(\sqrt{\frac{\log n}{n}}\right),$$

where $\tau(n)$ is the n-dimensional kissing number. This improves the standard greedy bound $M(n) \leq \tau(n)+1$ by a polynomial factor $O(\sqrt{n/\log n}) \to \infty$. The key inputs are: the geometric fact that the clique number of any packing's tangency graph satisfies $\omega \leq n+1$ (equidistant-set bound), the degree bound $\Delta \leq \tau(n)$, and the Bonamy–Kelly–Nelson–Postle theorem on coloring graphs with bounded maximum degree and small clique number.

Barrier. A systematic barrier theorem (Section 6) proves that the four classical spectral families from finite geometry (Kneser graphs, triangular graphs, symplectic polar graphs, and complements of hexagon

point graphs) all have Hoffman bound $H(G) \leq d$, so none can improve the lower-bound surplus beyond $O(d^{2/3})$ within this framework.

Problem

Throughout this paper, a *unit sphere* has radius $\frac{1}{2}$ (diameter 1). Two such spheres have disjoint interiors if and only if their centres are at distance ≥ 1; they are *tangent* when their centres are at distance exactly 1.

A *unit sphere packing* in $\mathbb{R}^n$ is a set of unit spheres (radius $\frac{1}{2}$) with pairwise disjoint interiors, equivalently a set of points (sphere centres) in $\mathbb{R}^n$ with pairwise distance ≥ 1. The *tangency graph* of the packing takes the centres as vertices and connects pairs at distance exactly 1 by an edge. A *colouring* assigns a colour to each sphere so that tangent pairs receive different colours; the *chromatic number* of the packing is the minimum number of colours needed.

Denote by $M(n)$ the supremum of chromatic numbers of all unit sphere packings (finite or infinite) in $\mathbb{R}^n$. The current state of knowledge is:

- *Upper bound*: $M(n) \leq \tau(n) + 1 \leq 2^{(0.401+o(1))n}$ by greedy colouring and the Kabatiansky–Levenshtein bound on the kissing number [4].
- *Lower bound*: $M(n) \geq n + \Omega(n^{2/3})$ for an infinite sequence of dimensions n, first proved by Hao Chen [1] via generalized quadrangles.

We provide: (a) a clean spectral proof of the lower bound establishing the explicit constant $\limsup_{n\to\infty}(M(n)-n)/n^{2/3} \geq 1$; (b) an improved upper bound $M(n) = O(\tau(n)\sqrt{\log n/n})$, a polynomial-factor improvement over greedy; and (c) a barrier theorem for the lower-bound framework.

Main Results

Section 1: Spectral Realization as a Unit Packing

Theorem 0.1 (Spectral Realization)**.** *Let $G = (V, E)$ be a finite connected non-complete k-regular graph on v vertices with adjacency matrix A, least eigenvalue $s < -1$ of multiplicity g, and $d = v - g - 1$. Then there exist points $\{p_x : x \in V\} \subset \mathbb{R}^d$ such that*

$$\|p_x - p_y\| = 1 \quad \text{if } xy \in E, \qquad \|p_x - p_y\| > 1 \quad \text{if } xy \notin E,\ x \neq y.$$

In particular, the tangency graph of the unit sphere packing $\{p_x\}$ (with minimum inter-centre distance 1*) is exactly G.*

We need to include one extra positive scale factor. The unscaled matrix PMP gives the right **relative** distances, but not generally unit edge length unless $s = -2$. We define the scaled Gram matrix accordingly.

Proof. Let $\mathbf{1} \in \mathbb{R}^v$ be the all-ones vector, let $J = \mathbf{1}\mathbf{1}^\mathsf{T}$, and let

$$P = I - \frac{1}{v} J.$$

Thus P is the orthogonal projection onto $\mathbf{1}^\perp$. Set

$$M = \frac{1}{-s}(A - sI), \qquad \alpha = \frac{s}{2(s+1)}.$$

Since $s < -1$, we have $\alpha > 0$. We will use

$$B = \alpha PMP$$

as the Gram matrix.

Because s is the least eigenvalue of the symmetric matrix A, every eigenvalue of $A - sI$ is nonnegative. Since $-s > 0$, it follows that $M \succeq 0$. Hence for every $z \in \mathbb{R}^v$,

$$z^\mathsf{T} PMPz = (Pz)^\mathsf{T} M(Pz) \geq 0,$$

so $PMP \succeq 0$, and therefore $B \succeq 0$.

Rank computation. Since G is k-regular, $A\mathbf{1} = k\mathbf{1}$. Moreover, because G is connected, the k-eigenspace is exactly $\operatorname{span}\{\mathbf{1}\}$. (Proof: if $Au = ku$ and $u_x := u(x)$ achieves its maximum at some vertex x_0, then $ku_{x_0} = \sum_{y \sim x_0} u_y \leq ku_{x_0}$ with equality only if all neighbours of x_0 also achieve the maximum; by connectedness all entries of u equal u_{x_0}, so $u \in \operatorname{span}\{\mathbf{1}\}$.)

Take an orthogonal eigenbasis of A. The vector $\mathbf{1}$ contributes one eigenvector with eigenvalue k. The eigenspace for the least eigenvalue s has dimension g and lies inside $\mathbf{1}^\perp$ (because for any eigenvector u with $Au = su$ and $s \neq k$, we have $k(1^T u) = (k\mathbf{1})^T u = (A\mathbf{1})^T u = \mathbf{1}^T Au = s(\mathbf{1}^T u)$, so $(k - s)(\mathbf{1}^T u) = 0$, hence $\mathbf{1}^T u = 0$). On $\mathbf{1}^\perp$, the projection P acts as the identity, so if $u \perp \mathbf{1}$ and $Au = \lambda u$, then

$$PMPu = Mu = \frac{\lambda - s}{-s} u.$$

This eigenvalue is 0 exactly when $\lambda = s$, and is positive when $\lambda > s$. Also, $PMP\mathbf{1} = 0$. Therefore,

$$\ker(PMP) = \operatorname{span}\{\mathbf{1}\} \oplus \ker(A - sI),$$

so $\dim \ker(PMP) = 1 + g$, and

$$\operatorname{rank}(B) = \operatorname{rank}(PMP) = v - (1 + g) = v - g - 1 = d.$$

Gram factorization. Since B is a real positive-semidefinite matrix of rank d, it admits a Gram factorisation: let $B = U\Delta U^\mathsf{T}$ be its eigendecomposition, where $\Delta = \operatorname{diag}(\beta_1, \ldots, \beta_d, 0, \ldots, 0)$ with $\beta_1, \ldots, \beta_d > 0$ the positive eigenvalues, and U orthogonal. Set $F = [\sqrt{\beta_1}\, u_1 \mid \cdots \mid \sqrt{\beta_d}\, u_d] \in \mathbb{R}^{v \times d}$, so that $FF^\mathsf{T} = \sum_{i=1}^d \beta_i u_i u_i^\mathsf{T} = B$. For each vertex $x \in V$, define $p_x \in \mathbb{R}^d$ to be the x-th row of F. Then $B_{xy} = \langle p_x, p_y \rangle$.

Distance computation. Let

$$\rho = \frac{k-s}{-s}.$$

Every row sum of M equals ρ: since $\sum_y A_{xy} = k$ (regularity) and $\sum_y I_{xy} = 1$,

$$\sum_y M_{xy} = \frac{1}{-s}(k-s) = \rho.$$

Since M is symmetric, every column sum also equals ρ. Therefore $M\mathbf{1} = \rho\mathbf{1}$, giving $MJ = \rho J$ and $JM = \rho J$, so $JMJ = J(\rho\mathbf{1}\mathbf{1}^T) = \rho v J$. Expanding:

$$PMP = \left(I - \tfrac{1}{v}J\right)M\left(I-\tfrac{1}{v}J\right) = M - \tfrac{1}{v}MJ - \tfrac{1}{v}JM + \tfrac{1}{v^2}JMJ = M - \tfrac{\rho}{v}J - \tfrac{\rho}{v}J + \tfrac{\rho}{v}J = M - \frac{\rho}{v}J.$$

For every vertex x, $M_{xx} = \frac{-s}{-s} = 1$. For distinct vertices $x \neq y$,

$$M_{xy} = \begin{cases} \dfrac{1}{-s}, & xy \in E, \\ 0, & xy \notin E. \end{cases}$$

Therefore $(B_0)_{xx} = 1 - \frac{\rho}{v}$ (where $B_0 = PMP$), and for $x \neq y$,

$$(B_0)_{xy} = \begin{cases} -\dfrac{1}{s} - \dfrac{\rho}{v}, & xy \in E, \\ -\dfrac{\rho}{v}, & xy \notin E. \end{cases}$$

Now let $x \neq y$. Since $B = \alpha B_0$ is the Gram matrix of the p_x's,

$$\|p_x - p_y\|^2 = B_{xx} + B_{yy} - 2B_{xy}.$$

If $xy \in E$:

$$\|p_x - p_y\|^2 = \alpha\Big[2\Big(1 - \tfrac{\rho}{v}\Big) - 2\Big(-\tfrac{1}{s} - \tfrac{\rho}{v}\Big)\Big] = \alpha\Big(2 + \tfrac{2}{s}\Big) = \frac{s}{2(s+1)} \cdot \frac{2(s+1)}{s} = 1.$$

Thus $\|p_x - p_y\| = 1$ whenever $xy \in E$.

If $xy \notin E$, $x \neq y$:

$$\|p_x - p_y\|^2 = \alpha\Big[2\Big(1 - \tfrac{\rho}{v}\Big) - 2\Big(-\tfrac{\rho}{v}\Big)\Big] = 2\alpha = \frac{s}{s+1}.$$

Since $s < -1$, write $s = -1 - \varepsilon$ with $\varepsilon > 0$; then $\frac{s}{s+1} = \frac{1+\varepsilon}{\varepsilon} > 1$. Hence $\|p_x - p_y\| > 1$ whenever $xy \notin E$ and $x \neq y$.

Therefore the point set $\{p_x : x \in V\} \subset \mathbb{R}^d$ has minimum pairwise distance at least 1, with equality exactly on edges of G. Consequently, the tangency graph of this unit sphere packing is exactly G. $\square$

Section 2: Hoffman Lower Bound for Chromatic Number

Theorem 0.2 (Hoffman Bound)**.** *Let G be a finite k-regular graph on v vertices with least adjacency eigenvalue $s < 0$. Then*

$$\chi(G) \;\geq\; 1 - \frac{k}{s}.$$

Proof. For reference, the inequality to be proved is the standard Hoffman lower bound on the chromatic number: $\chi(G) \geq 1 - \lambda_1/\lambda_n$, where λ_1 and λ_n are the largest and smallest adjacency eigenvalues. We prove the stated k-regular case directly.

Let A be the adjacency matrix of G, and let $\mathbf{1} \in \mathbb{R}^v$ be the all-ones vector. Since G is k-regular, $A\mathbf{1} = k\mathbf{1}$.

Let $S \subseteq V(G)$ be any independent set, and let $\mathbf{x}$ be its indicator vector. Put $n := |S| = \mathbf{x}^T\mathbf{x}$. If $n = 0$, there is nothing to prove, so assume $n > 0$.

Decompose $\mathbf{x}$ into its component along $\mathbf{1}$ and its orthogonal component:

$$\mathbf{x} = \frac{n}{v}\mathbf{1} + \mathbf{u}, \qquad \mathbf{u} \perp \mathbf{1}.$$

Indeed, $\mathbf{1}^T\mathbf{u} = \mathbf{1}^T\mathbf{x} - \frac{n}{v}\mathbf{1}^T\mathbf{1} = n - \frac{n}{v}v = 0$.

Because A is real symmetric, it has an orthonormal eigenbasis. Every eigenvalue of A is at least s, so for every vector $\mathbf{u} \perp \mathbf{1}$ we have $\mathbf{u}^T A\mathbf{u} \geq s\|\mathbf{u}\|^2$.

Now compute $\mathbf{x}^T A\mathbf{x}$. Since S is independent, no edge has both endpoints in S, so $\mathbf{x}^T A\mathbf{x} = 0$. On the other hand,

$$\mathbf{x}^T A\mathbf{x} = \left(\frac{n}{v}\mathbf{1} + \mathbf{u}\right)^T A\left(\frac{n}{v}\mathbf{1} + \mathbf{u}\right) = \frac{n^2}{v^2}\mathbf{1}^T A\mathbf{1} + \frac{2n}{v}\mathbf{1}^T A\mathbf{u} + \mathbf{u}^T A\mathbf{u}.$$

Since $A\mathbf{1} = k\mathbf{1}$ and $\mathbf{u} \perp \mathbf{1}$, we have $\mathbf{1}^T A\mathbf{1} = kv$ and $\mathbf{1}^T A\mathbf{u} = (A\mathbf{1})^T\mathbf{u} = k\mathbf{1}^T\mathbf{u} = 0$. Therefore

$$0 = \mathbf{x}^T A\mathbf{x} = \frac{kn^2}{v} + \mathbf{u}^T A\mathbf{u}.$$

Using $\mathbf{u}^T A\mathbf{u} \geq s\|\mathbf{u}\|^2$ gives

$$0 \geq \frac{kn^2}{v} + s\|\mathbf{u}\|^2.$$

Now $\|\mathbf{u}\|^2 = \|\mathbf{x} - \frac{n}{v}\mathbf{1}\|^2 = \|\mathbf{x}\|^2 - \frac{n^2}{v} = n - \frac{n^2}{v} = \frac{n(v-n)}{v}$. Hence

$$0 \geq \frac{kn^2}{v} + s\frac{n(v-n)}{v}.$$

Multiplying by $\frac{v}{n} > 0$ yields

$$0 \geq kn + s(v-n) = (k-s)n + sv.$$

Thus $(k-s)n \leq -sv$. Since $s < 0$ and $k - s > 0$, we obtain

$$n \leq \frac{-sv}{k-s}.$$

So every independent set in G has size at most $\frac{-sv}{k-s}$.

Now let $V_1, \dots, V_\chi$ be the color classes of an optimal proper coloring of G, where $\chi = \chi(G)$. Each V_i is an independent set, so $|V_i| \le \frac{-sv}{k-s}$ for each i. Since the color classes partition $V(G)$,

$$v = \sum_{i=1}^{\chi} |V_i| \le \chi \cdot \frac{-sv}{k-s}.$$

Dividing by $v > 0$ gives $1 \le \chi \cdot \frac{-s}{k-s}$. Therefore

$$\chi \ge \frac{k-s}{-s} = 1 - \frac{k}{s}. \qquad \square$$

$\square$

Section 3: The Generalized Quadrangle Construction

Theorem 0.3 (Spectral Family from $\mathrm{GQ}(q, q^2)$)**.** *For every prime power $q \ge 2$, let G_q denote the complement of the point-collinearity graph of the generalized quadrangle* $\mathrm{GQ}(q, q^2)$*. Then:*

1. *G_q has $v_q = (q+1)(q^3+1)$ vertices and is regular of degree $k_q = q^4$.*
2. *The least adjacency eigenvalue of G_q is $s_q = -q$, with multiplicity $g_q = q^4 + q^2$, and the embedding dimension is $d_q := v_q - g_q - 1 = q^3 - q^2 + q$.*
3. *The Hoffman lower bound satisfies*

$$1 - \frac{k_q}{s_q} = q^3 + 1 = d_q + (q^2 - q + 1).$$

4. *For every prime power $q \ge 2$,*

$$1 - \frac{k_q}{s_q} \ge d_q + \left(\frac{3}{4}\right)^{1/3} d_q^{2/3}.$$

Proof. Let $\mathcal{Q}$ be a generalized quadrangle of order (q, q^2), and let Γ_q be its point-collinearity graph. Thus two distinct points are adjacent in Γ_q exactly when they are collinear in $\mathcal{Q}$. Generalized quadrangles of order (q, q^2) exist for all prime powers q: the classical realization is the *elliptic quadric* $Q^-(5, q)$ in $\mathrm{PG}(5, q)$, which has $(q^3+1)(q+1)$ points and forms a $\mathrm{GQ}(q, q^2)$ (see [5] and [2]). The Hermitian variety $H(3, q^2)$ gives $\mathrm{GQ}(q^2, q)$, not $\mathrm{GQ}(q, q^2)$.

The collinearity graph of a $\mathrm{GQ}(s, t)$ is strongly regular with parameters [5]

$$v = (s+1)(st+1), \quad k = s(t+1), \quad \lambda = s - 1, \quad \mu = t + 1,$$

and nontrivial eigenvalues $s-1$ (multiplicity $st(s+1)(t+1)/(s+t)$) and $-(t+1)$ (multiplicity $s^2(st+1)/(s+t)$).

Setting $s = q$ and $t = q^2$:

$$v = (q+1)(q^3+1), \quad \kappa = q(q^2+1), \quad \lambda = q-1, \quad \mu = q^2+1,$$

with nontrivial eigenvalues

$$r = q-1, \qquad s_0 = -(q^2+1).$$

Connectedness. If two points are collinear, they are adjacent in Γ_q. If two points are not collinear, they have exactly $\mu = q^2+1 > 0$ common neighbours. Hence every pair of vertices is at distance at most 2 in Γ_q, so Γ_q is connected. For any connected k-regular graph, the degree eigenvalue k has multiplicity 1: if $Au = \kappa u$ then the maximum-entry argument (same as in Section 1) shows $u \in \operatorname{span}\{\mathbf{1}\}$, so the κ-eigenspace is 1-dimensional.

Multiplicity computation. Let the multiplicities of r and s_0 be f and g_0 respectively. The trace of the adjacency matrix of a simple graph (no self-loops) equals 0 since $\operatorname{tr}(A) = \sum_x A_{xx} = 0$; by the spectral theorem, this equals the sum of eigenvalues with multiplicity: $\kappa \cdot 1 + r \cdot f + s_0 \cdot g_0 = 0$. Combined with $1 + f + g_0 = v$, the trace condition is $\kappa + (q-1)f - (q^2+1)g_0 = 0$. Using $g_0 = v - 1 - f$:

$$\kappa + (q-1)f - (q^2+1)(v-1-f) = 0 \implies (q^2+q)f = (q^2+1)(v-1) - \kappa.$$

Substituting $v - 1 = q^4 + q^3 + q$ and $\kappa = q(q^2+1)$:

$$f = \frac{(q^2+1)(q^4+q^3+q) - q(q^2+1)}{q^2+q} = \frac{(q^2+1)(q^4+q^3)}{q(q+1)} = q^2(q^2+1) = q^4+q^2.$$

Hence $g_0 = v - 1 - f = (q^4+q^3+q) - (q^4+q^2) = q^3 - q^2 + q$.

Complement eigenvalues. Let $G_q = \overline{\Gamma_q}$ with adjacency matrix $B = J - I - A$. Since Γ_q is κ-regular, $B\mathbf{1} = (v-1-\kappa)\mathbf{1}$, so G_q has degree

$$k_q = v - 1 - \kappa = (q^4+q^3+q) - (q^3+q) = q^4.$$

This proves item (1).

For the non-principal eigenvalues: if $Au = \theta u$ with $\theta \neq \kappa$, then $\kappa(\mathbf{1}^T u) = \mathbf{1}^T A u = \theta(\mathbf{1}^T u)$, so $\mathbf{1}^T u = 0$. Thus all non-principal eigenvectors lie in $\mathbf{1}^\perp$, and for $u \perp \mathbf{1}$ with $Au = \theta u$: $Bu = (J - I - A)u = 0 - u - \theta u = (-1-\theta)u$. Hence G_q has eigenvalues:

- q^4 (multiplicity 1, principal eigenvalue),
- $-r - 1 = -q$ (multiplicity $f = q^4 + q^2$),
- $-s_0 - 1 = q^2$ (multiplicity $g_0 = q^3 - q^2 + q$).

For $q \geq 2$, the three eigenvalues satisfy $q^4 > 0$, $q^2 > 0$, and $-q < 0$. Hence the least adjacency eigenvalue of G_q is $s_q = -q$ with multiplicity $g_q = f = q^4 + q^2$.

Embedding dimension.

$$d_q := v_q - g_q - 1 = (q^4 + q^3 + q + 1) - (q^4 + q^2) - 1 = q^3 - q^2 + q.$$

Non-completeness and connectivity of G_q. We have $k_q = q^4 < q^4 + q^3 + q = v_q - 1$, so G_q is not complete.

For connectivity, recall that two vertices x, y are *adjacent in G_q* if and only if they are *non-collinear in $\mathcal{Q}$*. We show every pair $x \neq y$ has a common neighbour in G_q, establishing diameter at most 2.

Case 1: x and y are adjacent in G_q (non-collinear in $\mathcal{Q}$). Then xy is already an edge, and they trivially have a common neighbour if we can find $z \neq x, y$ adjacent to both; the diameter-2 bound suffices for connectivity, but adjacency already gives a path.

Case 2: x and y are non-adjacent in G_q (collinear in $\mathcal{Q}$). We need a vertex z adjacent to both x and y in G_q, i.e., non-collinear with both. In $\mathcal{Q}$, each point has $\kappa = q(q^2+1)$ collinear points, so x is non-collinear with exactly $v-1-\kappa = q^4$ points of $\mathcal{Q} \setminus \{x\}$; call this set A. Similarly let B be the q^4 points non-collinear with y. Since x and y are collinear, $x \notin A$ and $y \notin B$ (a point is always collinear with itself, but the exclusion $\{x, y\}$ merely removes two points from the ambient set). More precisely, $A \subseteq V \setminus \{x\}$ and $B \subseteq V \setminus \{y\}$, and since x is collinear with y we have $x \notin B$ and $y \notin A$. So both A and B are subsets of $V \setminus \{x, y\}$, which has $v_q - 2 = q^4 + q^3 + q - 1$ elements.

Since $|A| + |B| = 2q^4 > q^4 + q^3 + q - 1 = |V \setminus \{x, y\}|$ for all $q \geq 2$ (because $q^4 - (q^3+q-1) = (q-1)(q^3-1) > 0$ for $q \geq 2$), by inclusion-exclusion $A \cap B \neq \emptyset$. Any $z \in A \cap B$ is non-collinear with both x and y, so z is adjacent to both in G_q, giving the path x-z-y.

Thus G_q has diameter at most 2 and is connected. This proves item (2).

Hoffman bound.

$$1 - \frac{k_q}{s_q} = 1 - \frac{q^4}{-q} = 1 + q^3 = q^3 + 1.$$

Since $d_q + (q^2 - q + 1) = (q^3 - q^2 + q) + (q^2 - q + 1) = q^3 + 1$, we have

$$1 - \frac{k_q}{s_q} = d_q + (q^2 - q + 1).$$

This proves item (3).

Explicit surplus bound. Let $A_q = q^2 - q + 1$, so $d_q = qA_q$ and the surplus equals A_q. It suffices to show $A_q \geq (3/4)^{1/3} d_q^{2/3} = (3/4)^{1/3}(qA_q)^{2/3}$. Since $A_q > 0$, cubing gives the equivalent inequality $A_q^3 \geq (3/4)q^2 A_q^2$, i.e. $A_q \geq (3/4)q^2$. Computing:

$$A_q - \frac{3}{4}q^2 = q^2 - q + 1 - \frac{3}{4}q^2 = \frac{1}{4}q^2 - q + 1 = \frac{(q-2)^2}{4} \geq 0.$$

Therefore $q^2 - q + 1 \geq (3/4)^{1/3} d_q^{2/3}$ for all q, and item (4) follows. $\square$

Section 4: Packing Lower Bounds at the Special Dimensions

Theorem 0.4 (Main Packing Bound)**.** *For every prime power $q \geq 2$, the supremum $M(d_q)$ of chromatic numbers of finite unit sphere packings in $\mathbb{R}^{d_q}$ satisfies*

$$M(d_q) \;\geq\; q^3 + 1 \;=\; d_q + (q^2 - q + 1) \;\geq\; d_q + \left(\frac{3}{4}\right)^{1/3} d_q^{2/3},$$

where $d_q = q^3 - q^2 + q$.

Proof. Fix a prime power $q \geq 2$. Let G_q be the graph from Section 3. We explicitly verify that G_q meets all the hypotheses of the Spectral Realization Theorem (Section 1):

- *Finite*: G_q has $v_q = (q+1)(q^3+1)$ vertices (Section 3, item 1).
- *Connected*: proved in Section 3 (diameter-at-most-2 argument via the $|A \cap B| \neq \emptyset$ counting for non-adjacent pairs).
- *Non-complete*: $k_q = q^4 < v_q - 1$ (Section 3, item 1).
- *k_q-regular*: G_q is regular of degree $k_q = q^4$ (Section 3, item 1).
- *Least eigenvalue $s_q = -q < -1$*: since $q \geq 2$, we have $s_q = -q \leq -2 < -1$ (Section 3, item 2).
- *Multiplicity g_q*: $g_q = q^4 + q^2$ (Section 3, item 2).
- *Dimension formula*: $d_q = v_q - g_q - 1 = q^3 - q^2 + q$ (Section 3, item 2).

By the Spectral Realization Theorem, there exist points $\{p_x : x \in V(G_q)\} \subset \mathbb{R}^{d_q}$ forming a unit sphere packing in $\mathbb{R}^{d_q}$ whose tangency graph is exactly G_q. Therefore

$$M(d_q) \geq \chi(G_q).$$

Since G_q is k_q-regular with least eigenvalue $s_q = -q < 0$, the Hoffman Bound (Section 2) gives

$$\chi(G_q) \geq 1 - \frac{k_q}{s_q} = 1 + q^3 = q^3 + 1.$$

By Section 3, item 3 and item 4:

$$q^3 + 1 = d_q + (q^2 - q + 1) \geq d_q + \left(\frac{3}{4}\right)^{1/3} d_q^{2/3}.$$

Combining: $M(d_q) \geq \chi(G_q) \geq q^3 + 1 \geq d_q + (3/4)^{1/3} d_q^{2/3}$. $\square$

Section 5: Monotonicity and the Limsup Lower Bound

Theorem 0.5 (Monotonicity and Limsup Bound)**.** *Let $M(n)$ denote the supremum of chromatic numbers of unit sphere packings (finite or infinite) in $\mathbb{R}^n$. Then:*

1. ***(Monotonicity)*** *$M(n+1) \geq M(n)$ for all $n \geq 1$.*
2. ***(Limsup lower bound)***

$$\limsup_{n\to\infty} \frac{M(n) - n}{n^{2/3}} \geq 1.$$

Remark 0.6. The constant 1 is approached but not necessarily achieved: the proof gives $(M(d_q) - d_q)/d_q^{2/3} \geq (1 - 1/q + 1/q^2)^{1/3}$, which tends to 1 from below as $q \to \infty$. No upper bound of the form $M(n) - n = O(n^{2/3})$ is claimed here.

Remark 0.7. The gap $d_{q+1} - d_q = 3q^2 + q + 1 \sim 3d_q^{2/3}$ is of the same order as the surplus $q^2 - q + 1 \sim d_q^{2/3}$, so a naive monotonicity interpolation does *not* extend the bound $M(n) \geq n + c\,n^{2/3}$ to all large n from this sequence alone.

Proof. **(1) Monotonicity.** Define the embedding $\iota : \mathbb{R}^n \to \mathbb{R}^{n+1}$ by $\iota(p) = (p, 0)$. For any unit sphere packing $\mathcal{P}$ in $\mathbb{R}^n$ (finite or infinite), the image $\iota(\mathcal{P}) = \{(p, 0) : p \in \mathcal{P}\}$ is a unit sphere packing in $\mathbb{R}^{n+1}$, because

$$\|\iota(p) - \iota(q)\|_{\mathbb{R}^{n+1}} = \|p - q\|_{\mathbb{R}^n}$$

for all $p, q \in \mathcal{P}$. In particular, distances are preserved, so tangencies are preserved, and the tangency graph of $\iota(\mathcal{P})$ is isomorphic to that of $\mathcal{P}$. Hence every chromatic number achievable by a packing in $\mathbb{R}^n$ is also achievable by a packing in $\mathbb{R}^{n+1}$, giving $M(n) \leq M(n+1)$.

(2) Limsup bound. By the Main Packing Bound (Section 4), for every prime power $q \geq 2$:

$$M(d_q) \;\geq\; q^3 + 1 \;=\; d_q + (q^2 - q + 1), \qquad d_q = q^3 - q^2 + q.$$

Since $d_q = q(q^2 - q + 1)$, we have $d_q^{2/3} = q^{2/3}(q^2 - q + 1)^{2/3}$, so

$$\frac{M(d_q) - d_q}{d_q^{2/3}} \;\geq\; \frac{q^2 - q + 1}{d_q^{2/3}} = \frac{(q^2 - q + 1)^{1/3}}{q^{2/3}} = \left(1 - \frac{1}{q} + \frac{1}{q^2}\right)^{1/3}.$$

The function $f(q) = 1 - 1/q + 1/q^2$ satisfies $f'(q) = 1/q^2 - 2/q^3 = (q-2)/q^3 \geq 0$ for $q \geq 2$, so f is nondecreasing for $q \geq 2$ and $\lim_{q\to\infty} f(q) = 1$.

Fix any $\varepsilon \in (0, 1)$. Since $f(q) \to 1$, there exists a prime power q_0 such that $f(q_0) > 1 - \varepsilon$. By Euclid's theorem on the infinitude of primes, there exist infinitely many prime powers $q > q_0$, giving infinitely many indices $n_j = d_{q_j} \to \infty$ with

$$\frac{M(n_j) - n_j}{n_j^{2/3}} \;\geq\; f(q_j)^{1/3} \;>\; (1 - \varepsilon)^{1/3}.$$

Therefore $\limsup_{n\to\infty}(M(n)-n)/n^{2/3} \geq (1-\varepsilon)^{1/3}$ for every $\varepsilon \in (0,1)$. Letting $\varepsilon \to 0$ gives

$$\limsup_{n\to\infty} \frac{M(n)-n}{n^{2/3}} \;\geq\; 1. \qquad \square$$

$\square$

Section 6: Barrier Theorem for the Hoffman Framework

The main result gives $M(n) \geq n+\Omega(n^{2/3})$ via the Spectral Realization Theorem and the Hoffman bound. A natural question is whether the same framework can yield a larger surplus of order n^α for some $\alpha > 2/3$. This section shows that for all standard graph families from finite geometry the answer is no.

We collect the framework parameters. For a connected k-regular graph G with v vertices and least eigenvalue $s < -1$ of multiplicity g:

- embedding dimension: $d = v - g - 1$,
- Hoffman lower bound: $H(G) := 1 - k/s$,
- surplus: $H(G) - d$.

Remark 0.8. Throughout this section, "standard spectral families from finite geometry" refers to exactly the four families listed below: Kneser graphs, triangular graphs, non-collinearity graphs of symplectic polar spaces, and complements of generalized-hexagon point graphs. The theorem makes no claim about other families.

Theorem 0.9 (Hoffman-Framework Barrier). *For the following four families, the Hoffman bound does not exceed d:*

1. (Kneser $K(n,k)$, fixed $k \geq 3$): *$H = n/k$, $d \sim n^k/k!$, so $H = o(d)$.*
2. (Triangular $T(n)$, $n \geq 5$): *$H = d = n-1$, surplus $= 0$.*
3. (Non-collinearity of $W(2r-1,q)$, $r \geq 2$): *$H \leq d$, equality only at $(r,q) = (2,2)$.*
4. (Hexagon complement, order (q,q)): *$H \sim \frac{1}{2}q^4$, $d \sim \frac{5}{6}q^5$, so $H = o(d)$.*

None of these families achieves $H(G) \geq d + c\,d^\alpha$ for any $c > 0$, $\alpha > 0$.

Proof. **(1) Kneser graph $K(n,k)$, fixed $k \geq 3$, $n > 2k$.** The Kneser graph $K(n,k)$ has vertex set $\binom{[n]}{k}$ (all k-subsets of $[n]$), adjacent when disjoint. For $n > 2k$, its adjacency eigenvalues are

$$\lambda_j = (-1)^j \binom{n-k-j}{k-j}, \quad 0 \leq j \leq k,$$

with multiplicities $\binom{n}{j} - \binom{n}{j-1}$ for $j \geq 1$ [2]. In particular the $j = 1$ term,

$$\lambda_1 = -\binom{n-k-1}{k-1},$$

has multiplicity $\binom{n}{1} - \binom{n}{0} = n - 1$.

Claim: λ_1 is the least eigenvalue for every $n > 2k$, $k \geq 3$.

Proof: The even-j eigenvalues are positive for $n > 2k$. For odd $j \geq 3$, compare with $j = 1$:

$$\frac{|\lambda_1|}{|\lambda_j|} = \frac{\binom{n-k-1}{k-1}}{\binom{n-k-j}{k-j}} = \prod_{i=1}^{j-1} \frac{n-k-i}{k-i}.$$

For each $i \in \{1, \ldots, j-1\}$: since $j \leq k$ implies $i \leq k-2$, the denominator $k - i \geq 2 > 0$. Since $n > 2k$, we have $n - k - i > k - i$ (equivalently $n > 2k$), so each factor exceeds 1. Hence the product exceeds 1, giving $|\lambda_j| < |\lambda_1|$ for every odd $j \geq 3$. Thus $s_G = \lambda_1$ with $g = n - 1$. $\square$

The degree is $k_G = \binom{n-k}{k}$, so the Hoffman bound $H = 1 + \binom{n-k}{k}/\binom{n-k-1}{k-1} = n/k$, and $d = \binom{n}{k} - n \sim n^k/k!$. For fixed $k \geq 3$: $H/d \to 0$.

(2) Triangular graph $T(n)$, $n \geq 5$. The triangular graph $T(n)$ has vertex set $\binom{[n]}{2}$ with two vertices adjacent when they share one element; $v = \binom{n}{2}$, $k_G = 2(n-2)$. Two distinct 2-subsets of $[n]$ either are disjoint or share one element, so $T(n) = \overline{K(n,2)}$.

We derive the spectrum of $T(n)$ from the formula of Part (1) applied to $K(n,2)$ (with $k = 2$ [2]):

- $j = 1$: $\lambda_1 = -\binom{n-3}{1} = -(n-3)$, multiplicity $n - 1$.
- $j = 2$: $\lambda_2 = \binom{n-4}{0} = 1$, multiplicity $\binom{n}{2} - n$.

Each nontrivial eigenvector of $K(n,2)$ lies in $\mathbf{1}^\perp$ and becomes an eigenvector of $T(n) = \overline{K(n,2)}$ with eigenvalue $-1 - \lambda$, giving:

$$s_G = -1 - 1 = -2 \text{ (mult } g = \tbinom{n}{2} - n), \qquad n - 4 \text{ (mult } n - 1).$$

Hence $d = \binom{n}{2} - (\binom{n}{2} - n) - 1 = n - 1$ and $H = 1 - 2(n-2)/(-2) = n - 1 = d$; surplus $= 0$.

(3) Non-collinearity graph of $W(2r-1, q)$, $r \geq 2$. The symplectic polar space $W(2r-1, q)$ has $v = (q^{2r} - 1)/(q - 1)$ points. Its non-collinearity graph G is strongly regular with degree $k_G = q^{2r-1}$ and nontrivial eigenvalues $\pm q^{r-1}$ [2]. From the trace equations $f + g = v - 1$ and $f - g = -k_G/q^{r-1} = -q^r$:

$$g = \tfrac{v-1+q^r}{2}, \quad d = v - g - 1 = \tfrac{v-1-q^r}{2}, \quad H = 1 + q^r.$$

We show $d \geq H$, i.e. $v \geq 3 + 3q^r$, by three cases.

Case A: $r = 2$. $v = q^3 + q^2 + q + 1$, so $v - 3 - 3q^2 = q^3 - 2q^2 + q - 2 = (q-2)(q^2+1) \geq 0$, with equality iff $q = 2$. At $(r,q) = (2,2)$: $d = H = 5$.

Case B: $r = 3$, $q = 2$. Direct computation: $v = (2^6 - 1)/(2-1) = 63$, $d = (63-1-8)/2 = 27$, $H = 1+8 = 9$; $d = 27 > H = 9$.

Case C: $r \geq 3$, $(r,q) \neq (3,2)$. Here $q^{r-1} > 4$: if $r = 3$ then $q \geq 3$ gives $q^2 \geq 9$; if $r \geq 4$ then $q^{r-1} \geq 2^3 = 8$. Using $v \geq q^{2r-1}$ and $3 + 3q^r \leq 4q^r$:

$$v - 3 - 3q^r \geq q^{2r-1} - 4q^r = q^r(q^{r-1} - 4) > 0.$$

Thus $d > H$ in all cases except $(r,q) = (2,2)$ where $d = H$.

(4) Complement of the generalized hexagon point graph, order (q,q). Let Γ be the point-collinearity graph of a generalized hexagon of order (q,q) [2]: $v = (q+1)(q^4+q^2+1)$, degree $q(q+1)$, nontrivial eigenvalues $2q-1$ (mult m_1), -1 (mult m_2), $-(q+1)$ (mult m_3), where

$$m_1 = \tfrac{q(q+1)^2(q^2+q+1)}{6}, \quad m_2 = \tfrac{q(q+1)^2(q^2-q+1)}{2}, \quad m_3 = \tfrac{q(q^2+q+1)(q^2-q+1)}{3}.$$

For $G = \overline{\Gamma}$: $k_G = q^3(q^2+q+1)$, and the nontrivial eigenvalues of G are $-2q$ (mult m_1), 0 (mult m_2), q (mult m_3). The least eigenvalue is $s_G = -2q$ with multiplicity $g = m_1$, so

$$d = v - m_1 - 1 = m_2 + m_3 \sim \tfrac{5}{6}q^5, \qquad H = 1 + \tfrac{q^2(q^2+q+1)}{2} \sim \tfrac{1}{2}q^4.$$

Thus $H/d \sim 3/(5q) \to 0$. Explicit checks: for $q = 2$, $(m_1, m_2, m_3) = (21, 27, 14)$, $v = 63$, $d = 41$, $H = 15$; for $q = 3$, $d = 259$, $H = 59.5$. In all cases $H < d$.

Summary. For Kneser ($k \geq 3$) and the hexagon complement, $H = o(d)$; for the triangular graph and symplectic cases, $H \leq d$ with equality only at $(r,q) = (2,2)$. None achieves $H \geq d + c\,d^\alpha$ for any $c, \alpha > 0$. □

Remark 0.10. Any improvement to the surplus exponent $\alpha = 2/3$ within the Hoffman framework requires either a genuinely new graph family or a proof technique beyond the Hoffman ratio bound.

Section 7: Improved Upper Bound via Small-Clique Graph Coloring

The following theorem improves the standard upper bound $M(n) \leq \tau(n)+1$ by a polynomial factor, where $\tau(n)$ denotes the n-dimensional kissing number.

Theorem 0.11 (Improved Upper Bound)**.** *There exists an absolute constant $C > 0$ such that, for all sufficiently large n,*

$$M(n) \;\leq\; C\,\tau(n)\sqrt{\frac{\ln(n+2)}{\ln \tau(n)}}.$$

In particular, using the Kabatiansky–Levenshtein bound $\tau(n) \leq 2^{0.401n(1+o(1))}$,

$$M(n) \;=\; O\!\left(\tau(n)\cdot\sqrt{\frac{\log n}{n}}\right) \;=\; 2^{0.401n(1+o(1))}\cdot O\!\left(\sqrt{\frac{\log n}{n}}\right).$$

This is a polynomial-factor improvement over the greedy bound $M(n) \leq \tau(n)+1$.

The proof rests on three ingredients, which we state as lemmas.

Lemma 0.12 (de Bruijn–Erdős Compactness). *$M(n) = \sup\{\chi(G_X) : X \subset \mathbb{R}^n$ finite, $\|x-y\| \geq 1\ \forall x \neq y \in X\}$. In other words, $M(n)$ is achieved as a supremum over* finite *packings.*

Proof. Let $\mathcal{P}$ be any unit sphere packing in $\mathbb{R}^n$ (possibly infinite), and let $G = G_{\mathcal{P}}$ be its tangency graph. Every finite subgraph of G is the tangency graph of a finite sub-packing, hence has chromatic number at most $\sup_{\text{finite } X} \chi(G_X) =: K$. By the de Bruijn–Erdős theorem [6], an infinite graph whose every finite subgraph is K-colorable is itself K-colorable, so $\chi(G) \leq K$. Hence $M(n) \leq K$; the reverse inequality $M(n) \geq K$ is trivial. ☐

Lemma 0.13 (Geometric Graph Bounds). *For every finite unit sphere packing $X \subset \mathbb{R}^n$ (centres with pairwise distance ≥ 1), the tangency graph G_X satisfies:*

(a) (Degree bound) $\Delta(G_X) \leq \tau(n)$*, the n-dimensional kissing number.*

(b) (Clique bound) $\omega(G_X) \leq n+1$.

Proof. **(a).** The neighbours of a centre $v \in X$ in G_X are the centres $u \in X$ with $\|u - v\| = 1$, all lying on the unit sphere $S^{n-1}(v)$ of radius 1 centred at v, and pairwise at distance ≥ 1. Translating to $v = 0$, this is a set of unit-sphere points with pairwise angular separation ≥ 60ř, i.e. a kissing configuration. By definition of $\tau(n)$, the size of any kissing configuration is at most $\tau(n)$.

(b). A k-clique in G_X is a set $\{x_1, \ldots, x_k\} \subset X$ with $\|x_i - x_j\| = 1$ for all $i \neq j$ — an equidistant set of diameter 1 in $\mathbb{R}^n$. Setting $y_i = x_i - x_1$ for $i \geq 2$, we have $\|y_i\| = 1$ and

$$\|y_i - y_j\|^2 = \|x_i - x_j\|^2 = 1, \quad i \neq j \geq 2,$$

which gives $\langle y_i, y_j \rangle = \frac{1}{2}$ for all $i \neq j \geq 2$. The Gram matrix M of $y_2, \ldots, y_k$ satisfies $M_{ii} = 1$ and $M_{ij} = \frac{1}{2}$ for $i \neq j$, i.e. $M = \frac{1}{2}I + \frac{1}{2}J$ where J is the $(k-1) \times (k-1)$ all-ones matrix. The eigenvalues of M are $\frac{k}{2}$ (once) and $\frac{1}{2}$ ($k-2$ times), all positive, so M is positive definite and has rank $k-1$. Since $y_2, \ldots, y_k \in \mathbb{R}^n$, we need $k - 1 \leq n$, i.e. $k \leq n+1$. ☐

Lemma 0.14 (Bonamy–Kelly–Nelson–Postle, 2022 [7]). *There exists an absolute constant $C_0 > 0$ such that the following holds. For all $\omega \geq 3$ and all $\Delta \geq \omega^{C_0}$, every graph G with $\Delta(G) \leq \Delta$ and $\omega(G) \leq \omega$ satisfies*

$$\chi(G) \ \leq\ 72\,\Delta\sqrt{\frac{\ln \omega}{\ln \Delta}}.$$

Proof of Theorem 0.11. By Lemma 0.12, $M(n) = \sup_{\text{finite } X} \chi(G_X)$. Fix a finite unit sphere packing $X \subset \mathbb{R}^n$. By Lemma 0.13, G_X has $\Delta(G_X) \leq \tau(n)$ and $\omega(G_X) \leq n+1$.

We verify the hypothesis of Lemma 0.14 with $\Delta = \tau(n)$ and $\omega = n+2$: since $\tau(n) \geq 2^{0.2075n(1+o(1))}$ by the Kabatiansky–Levenshtein lower bound [4], we have $\ln \tau(n) \geq 0.2075n \ln 2$, while $\ln(n+2)^{C_0} = O(n)$, so $\tau(n) \geq (n+2)^{C_0}$ for all sufficiently large n. Hence

$$\chi(G_X) \;\leq\; 72\,\tau(n)\sqrt{\frac{\ln(n+2)}{\ln \tau(n)}}.$$

Taking the supremum over all finite X gives the first bound.

For the asymptotic form, since $\ln \tau(n) \leq 0.401n \ln 2 + o(n)$ (upper KL bound [4]),

$$\sqrt{\frac{\ln(n+2)}{\ln \tau(n)}} = O\left(\sqrt{\frac{\log n}{n}}\right),$$

giving $M(n) = O(\tau(n)\sqrt{\log n/n}) = 2^{0.401n(1+o(1))} \cdot O(\sqrt{\log n/n})$.

To see this is an improvement over greedy: the ratio of the new bound to $\tau(n)$ is $O(\sqrt{\log n/n}) \to 0$, so the new bound is $o(\tau(n))$ and hence $o(M_{\mathrm{greedy}}(n))$. $\square$

Remark 0.15. The key geometric input is $\omega(G_X) \leq n+1$, which contrasts sharply with $\tau(n) \sim 2^{0.401n}$. The logarithmic ratio $\ln \omega / \ln \Delta \sim (\log n)/(0.401n) \to 0$ is what drives the improvement. The de Bruijn–Erdős step (Lemma 0.12) is essential: the Bonamy–Kelly–Nelson–Postle theorem is a finite-graph result, and the transfer to arbitrary (possibly infinite) packings requires the compactness principle.

Corollary 0.16 (Main Result)**.** *The supremum $M(n)$ of chromatic numbers of unit sphere packings in $\mathbb{R}^n$ satisfies:*

(i) ***Lower bound:***

$$\limsup_{n\to\infty} \frac{M(n)-n}{n^{2/3}} \;\geq\; 1.$$

In particular, $(M(n)-n)/n^{2/3}$ is at least $(1-1/q+1/q^2)^{1/3} \to 1$ along $n = d_q = q^3 - q^2 + q$ as $q \to \infty$ through prime powers.

(ii) ***Upper bound:***

$$M(n) \;=\; O\left(\tau(n) \cdot \sqrt{\frac{\log n}{n}}\right) \;=\; 2^{0.401n(1+o(1))} \cdot O\left(\sqrt{\frac{\log n}{n}}\right).$$

(iii) ***Barrier:*** *By the Barrier Theorem* (Theorem 0.9)*, none of the four named classical spectral families from finite geometry (Kneser, triangular, symplectic polar, hexagon complement) achieves a positive surplus $H(G)-d > 0$ within the Hoffman framework.*

Proof. **Lower bound chain.** By the Spectral Realization Theorem (Theorem 0.1) and the explicit spectral computation in Theorem 0.3, the complement G_q of the collinearity graph of $\mathrm{GQ}(q, q^2)$ embeds as the tangency graph of a unit sphere packing in $\mathbb{R}^{d_q}$, so

$$M(d_q) \;\geq\; \chi(G_q).$$

By the Hoffman Bound (Theorem 0.2) and the eigenvalues $k_q = q^4$, $s_q = -q$ of G_q,

$$\chi(G_q) \;\geq\; 1 - \frac{k_q}{s_q} = q^3 + 1.$$

Hence $M(d_q) \geq q^3 + 1 = d_q + (q^2 - q + 1)$ (Theorem 0.4).

Limsup. By Theorem 0.5(2), taking $q \to \infty$ through prime powers,

$$\frac{M(d_q) - d_q}{d_q^{2/3}} \;\geq\; \left(1 - \frac{1}{q} + \frac{1}{q^2}\right)^{1/3} \;\to\; 1,$$

which gives $\limsup_{n\to\infty}(M(n) - n)/n^{2/3} \geq 1$.

Upper bound. This is Theorem 0.11.

Barrier. This is Theorem 0.9. $\square$

References

[1] H. Chen, *Ball packings with high chromatic numbers from strongly regular graphs*, Discrete Mathematics **340** (2017), no. 7. arXiv:1502.02070.

[2] A. E. Brouwer and H. Van Maldeghem, *Strongly Regular Graphs*, Cambridge University Press, 2022.

[3] A. J. Hoffman, *On eigenvalues and colorings of graphs*, in Graph Theory and its Applications (B. Harris, ed.), Academic Press, 1970, pp. 79–91.

[4] G. A. Kabatiansky and V. I. Levenshtein, *Bounds for packings on the sphere and in space*, Problemy Peredachi Informatsii **14** (1978), 3–25.

[5] S. E. Payne and J. A. Thas, *Finite Generalized Quadrangles*, Research Notes in Mathematics **110**, Pitman, Boston, 1984. Second edition: EMS Series of Lectures in Mathematics, European Mathematical Society, Zürich, 2009.

[6] N. G. de Bruijn and P. Erdős, *A colour problem for infinite graphs and a problem in the theory of relations*, Indagationes Mathematicae **13** (1951), 369–373.

[7] M. Bonamy, T. Kelly, P. Nelson, and L. Postle, *Bounding χ by a fraction of Δ for graphs without large cliques*, Journal of the European Mathematical Society **26** (2024), no. 5, 1451–1499. arXiv:1803.01051.